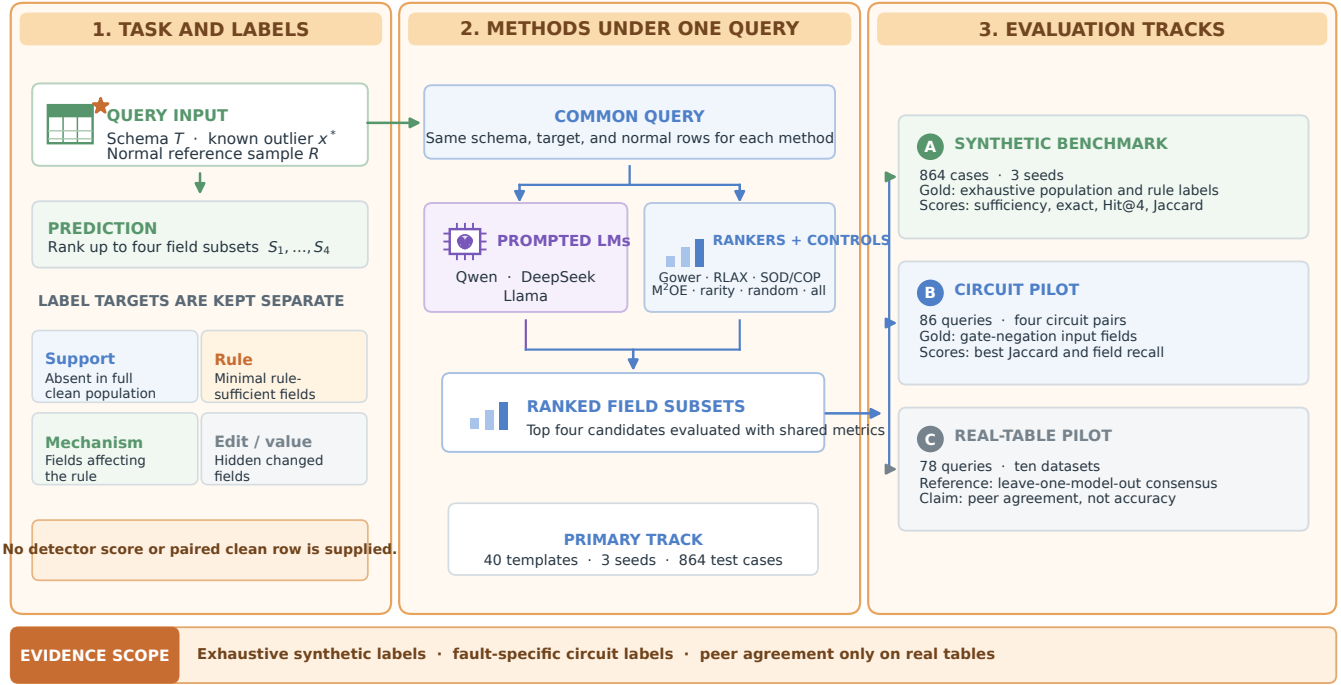
1. TASK AND LABELS
QUERY INPUT
Schema T · known outlier x^*
Normal reference sample R
PREDICTION
Rank up to four field subsets $S_1, \ldots, S_4$
LABEL TARGETS ARE KEPT SEPARATE
Support
Absent in full clean population
Rule
Minimal rule-sufficient fields
Mechanism
Fields affecting the rule
Edit / value
Hidden changed fields
No detector score or paired clean row is supplied.
2. METHODS UNDER ONE QUERY
COMMON QUERY
Same schema, target, and normal rows for each method
PROMPTED LMs
Qwen · DeepSeek
Llama
RANKERS + CONTROLS
Gower · RLAX · SOD/COP
M²OE · rarity · random · all
RANKED FIELD SUBSETS
Top four candidates evaluated with shared metrics
PRIMARY TRACK
40 templates · 3 seeds · 864 test cases
3. EVALUATION TRACKS
A
SYNTHETIC BENCHMARK
864 cases · 3 seeds
Gold: exhaustive population and rule labels
Scores: sufficiency, exact, Hit@4, Jaccard
B
CIRCUIT PILOT
86 queries · four circuit pairs
Gold: gate-negation input fields
Scores: best Jaccard and field recall
C
REAL-TABLE PILOT
78 queries · ten datasets
Reference: leave-one-model-out consensus
Claim: peer agreement, not accuracy
EVIDENCE SCOPE
Exhaustive synthetic labels · fault-specific circuit labels · peer agreement only on real tables